

Unit 10. Idempotent Matrices and Quadratic Forms

1 Idempotent matrices

1.1 Range, null space, and canonical form

Definition 10.1 (Idempotent matrix). A matrix $\mathbf{P} \in \mathbb{R}^{n \times n}$ is **idempotent** if

$$\mathbf{P}^2 = \mathbf{P}.$$

The definition includes the unique 0×0 matrix.

Origin: H §10.1; A&M §8.3

Theorem 10.2 (Range–null decomposition). For an idempotent $\mathbf{P} \in \mathbb{R}^{n \times n}$,

$$\begin{aligned}\mathbb{R}^n &= \text{col}(\mathbf{P}) \oplus \text{null}(\mathbf{P}), \\ \text{col}(\mathbf{P}) &= \text{null}(\mathbf{I}_n - \mathbf{P}), \\ \text{null}(\mathbf{P}) &= \text{col}(\mathbf{I}_n - \mathbf{P}).\end{aligned}$$

For every $\mathbf{x} \in \mathbb{R}^n$, the corresponding unique decomposition is

$$\mathbf{x} = \mathbf{P}\mathbf{x} + (\mathbf{I}_n - \mathbf{P})\mathbf{x}.$$

Origin: H Thm. 11.7.1 and Cor. 11.7.2

Proof. For every $\mathbf{x}$,

$$\mathbf{P}(\mathbf{P}\mathbf{x}) = \mathbf{P}\mathbf{x},$$

$$\mathbf{P}(\mathbf{I}_n - \mathbf{P})\mathbf{x} = (\mathbf{P} - \mathbf{P}^2)\mathbf{x} = \mathbf{0}.$$

Hence $\mathbf{P}\mathbf{x} \in \text{col}(\mathbf{P})$ and $(\mathbf{I}_n - \mathbf{P})\mathbf{x} \in \text{null}(\mathbf{P})$. If $\mathbf{z} \in \text{col}(\mathbf{P}) \cap \text{null}(\mathbf{P})$, then

$$\mathbf{z} = \mathbf{P}\mathbf{y}, \quad \mathbf{0} = \mathbf{P}\mathbf{z} = \mathbf{P}^2\mathbf{y} = \mathbf{P}\mathbf{y} = \mathbf{z}.$$

Thus $\mathbb{R}^n = \text{col}(\mathbf{P}) \oplus \text{null}(\mathbf{P})$.

Moreover,

$$\mathbf{x} \in \text{col}(\mathbf{P}) \iff \mathbf{P}\mathbf{x} = \mathbf{x} \iff (\mathbf{I}_n - \mathbf{P})\mathbf{x} = \mathbf{0},$$

$$\mathbf{x} \in \text{null}(\mathbf{P}) \iff \mathbf{x} = (\mathbf{I}_n - \mathbf{P})\mathbf{x} \iff \mathbf{x} \in \text{col}(\mathbf{I}_n - \mathbf{P}).$$

The same identities are valid when $n = 0$. □

Definition 10.3 (Projection along a complementary subspace). Let $\mathcal{S}, \mathcal{T} \subseteq \mathbb{R}^n$ satisfy $\mathbb{R}^n = \mathcal{S} \oplus \mathcal{T}$. The **projection onto $\mathcal{S}$ along $\mathcal{T}$** is the linear map

$$\mathbf{P}(\mathbf{s} + \mathbf{t}) = \mathbf{s}, \quad \mathbf{s} \in \mathcal{S}, \quad \mathbf{t} \in \mathcal{T}.$$

It is **oblique** when $\mathcal{T} \neq \mathcal{S}^\perp$.

Origin: H §14.12, p. 336

Proposition 10.4 (Idempotents are projections). A linear map $\mathbf{P} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is idempotent if and only if it is the projection onto $\text{col}(\mathbf{P})$ along $\text{null}(\mathbf{P})$. Conversely, the projection onto $\mathcal{S}$ along $\mathcal{T}$ is idempotent, has range $\mathcal{S}$, and has null space $\mathcal{T}$.

Origin: H Thm. 11.7.1

Proof. If $\mathbf{P}^2 = \mathbf{P}$, Theorem 10.2 gives

$$\mathbf{P}(\mathbf{s} + \mathbf{t}) = \mathbf{P}\mathbf{s} + \mathbf{P}\mathbf{t} = \mathbf{s}, \quad \mathbf{s} \in \text{col}(\mathbf{P}), \quad \mathbf{t} \in \text{null}(\mathbf{P}).$$

Conversely, for $\mathbf{x} = \mathbf{s} + \mathbf{t}$,

$$\mathbf{P}^2\mathbf{x} = \mathbf{P}\mathbf{s} = \mathbf{s} = \mathbf{P}\mathbf{x}.$$

The range and null assertions follow from $\mathbf{P}(\mathbf{s} + \mathbf{t}) = \mathbf{s}$ and the uniqueness of the direct-sum decomposition. $\square$

Theorem 10.5 (Canonical similarity). Let $\mathbf{P} \in \mathbb{R}^{n \times n}$ be idempotent and $r = \text{rank}(\mathbf{P})$. There is a nonsingular $\mathbf{T} \in \mathbb{R}^{n \times n}$ such that

$$\mathbf{T}^{-1}\mathbf{P}\mathbf{T} = \begin{pmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix}.$$

Conversely, every matrix similar to this block matrix is idempotent. The empty diagonal blocks cover $r = 0$ and $r = n$.

Origin: A&M Ex. 8.60; H §10.2, p. 189

Proof. Choose bases $\mathbf{s}_1, \dots, \mathbf{s}_r$ of $\text{col}(\mathbf{P})$ and $\mathbf{t}_1, \dots, \mathbf{t}_{n-r}$ of $\text{null}(\mathbf{P})$, and set

$$\mathbf{T} = (\mathbf{s}_1, \dots, \mathbf{s}_r, \mathbf{t}_1, \dots, \mathbf{t}_{n-r}), \quad \mathbf{D} = \text{diag}(\mathbf{I}_r, \mathbf{O}).$$

Theorem 10.2 gives $\mathbf{P}\mathbf{T} = \mathbf{T}\mathbf{D}$; hence $\mathbf{T}^{-1}\mathbf{P}\mathbf{T} = \mathbf{D}$. Conversely,

$$(\mathbf{T}\mathbf{D}\mathbf{T}^{-1})^2 = \mathbf{T}\mathbf{D}^2\mathbf{T}^{-1} = \mathbf{T}\mathbf{D}\mathbf{T}^{-1}.$$



Corollary 10.6 (Spectrum, trace, and rank). If $\mathbf{P} \in \mathbb{R}^{n \times n}$ is idempotent of rank r , then

$$\text{spec}(\mathbf{P}) \subseteq \{0, 1\}, \quad \text{tr}(\mathbf{P}) = \text{rank}(\mathbf{P}) = r, \quad \text{rank}(\mathbf{I}_n - \mathbf{P}) = n - r.$$

The matrix $\mathbf{P}$ is diagonalizable over $\mathbb{R}$.

Origin: H Cor. 10.2.2 and Lemma 10.2.4; A&M Exs. 8.56 and 8.61

Proof. Similarity preserves characteristic polynomials, trace, and rank. Apply Theorem 10.5 to $\text{diag}(\mathbf{I}_r, \mathbf{O})$ and to its complement $\text{diag}(\mathbf{O}, \mathbf{I}_{n-r})$. 

Example 10.1 (An oblique projection). For $a \in \mathbb{R}$,

$$\mathbf{P}_a = \begin{pmatrix} 1 & a \\ 0 & 0 \end{pmatrix}, \quad \mathbf{P}_a^2 = \mathbf{P}_a, \quad \text{col}(\mathbf{P}_a) = \text{span}\{\mathbf{e}_1\}, \quad \text{null}(\mathbf{P}_a) = \text{span}\{(-a, 1)'\}.$$

For $a \neq 0$, $\text{null}(\mathbf{P}_a) \neq \text{col}(\mathbf{P}_a)^\perp$ and $\mathbf{P}_a' \neq \mathbf{P}_a$.

Origin: Adapted from A&M Ex. 8.55

1.2 Orthogonal and complementary projections

Theorem 10.7 (Orthogonal versus oblique). For an idempotent $\mathbf{P} \in \mathbb{R}^{n \times n}$, the following are equivalent:

- (i) $\mathbf{P}' = \mathbf{P}$;
- (ii) $\text{col}(\mathbf{P}) \perp \text{null}(\mathbf{P})$;
- (iii) $\text{null}(\mathbf{P}) = \text{col}(\mathbf{P})^\perp$;
- (iv) $\mathbf{P}$ is the orthogonal projector onto $\text{col}(\mathbf{P})$.

Origin: H Thm. 12.3.9; A&M §8.3, p. 202

Proof. (i) $\Rightarrow$ (ii): if $\mathbf{s} = \mathbf{P}\mathbf{x}$ and $\mathbf{P}\mathbf{t} = \mathbf{0}$, then

$$\mathbf{s}'\mathbf{t} = \mathbf{x}'\mathbf{P}'\mathbf{t} = \mathbf{x}'\mathbf{P}\mathbf{t} = 0.$$

(ii) $\Rightarrow$ (iii): Theorem 10.2 and rank–nullity give

$$\text{null}(\mathbf{P}) \subseteq \text{col}(\mathbf{P})^\perp, \quad \dim \text{null}(\mathbf{P}) = n - r = \dim \text{col}(\mathbf{P})^\perp.$$

(iii) $\Rightarrow$ (iv) is Definition 10.3 with an orthogonal complement. For (iv) $\Rightarrow$ (i), let $\mathbf{x} = \mathbf{s} + \mathbf{t}$ and

$\mathbf{y} = \mathbf{u} + \mathbf{v}$, with $\mathbf{s}, \mathbf{u} \in \text{col}(\mathbf{P})$ and $\mathbf{t}, \mathbf{v} \in \text{col}(\mathbf{P})^\perp$. Then

$$\mathbf{x}'\mathbf{P}\mathbf{y} = (\mathbf{s} + \mathbf{t})'\mathbf{u} = \mathbf{s}'\mathbf{u} = \mathbf{s}'(\mathbf{u} + \mathbf{v}) = (\mathbf{P}\mathbf{x})'\mathbf{y}.$$

Thus $\mathbf{x}'(\mathbf{P} - \mathbf{P}')\mathbf{y} = 0$ for all $\mathbf{x}, \mathbf{y}$, so $\mathbf{P} = \mathbf{P}'$. □

Proposition 10.8 (Complement). The matrix $\mathbf{P}$ is idempotent if and only if $\mathbf{I}_n - \mathbf{P}$ is idempotent. In that case,

$$\mathbf{P}(\mathbf{I}_n - \mathbf{P}) = (\mathbf{I}_n - \mathbf{P})\mathbf{P} = \mathbf{O},$$

and the two matrices project along the exchanged decomposition

$$\mathbb{R}^n = \text{col}(\mathbf{P}) \oplus \text{null}(\mathbf{P}).$$

If $\mathbf{P}$ is symmetric, both projections are orthogonal.

Origin: H Lemma 10.1.2; A&M Ex. 8.53

Proof.

$$(\mathbf{I}_n - \mathbf{P})^2 - (\mathbf{I}_n - \mathbf{P}) = \mathbf{P}^2 - \mathbf{P}.$$

If this matrix is zero, then

$$\mathbf{P}(\mathbf{I}_n - \mathbf{P}) = \mathbf{P} - \mathbf{P}^2 = \mathbf{O} = (\mathbf{I}_n - \mathbf{P})\mathbf{P}.$$

The range and null spaces follow from Theorem 10.2; symmetry is preserved by complementation. $\square$

Corollary 10.9 (Symmetric idempotents are positive semidefinite). If $\mathbf{P}' = \mathbf{P} = \mathbf{P}^2$, then

$$\mathbf{x}'\mathbf{P}\mathbf{x} = \mathbf{x}'\mathbf{P}'\mathbf{P}\mathbf{x} = \|\mathbf{P}\mathbf{x}\|^2 \geq 0, \quad \mathbf{x}'(\mathbf{I}_n - \mathbf{P})\mathbf{x} = \|(\mathbf{I}_n - \mathbf{P})\mathbf{x}\|^2 \geq 0.$$

Thus $\mathbf{O} \preceq \mathbf{P} \preceq \mathbf{I}_n$.

Origin: H Lemma 14.2.17; A&M Ex. 8.66(a)

Proof. The displayed identities use symmetry and idempotency. The two inequalities are the definitions of positive semidefiniteness and the Löwner order. $\square$

1.3 Sums and products

Theorem 10.10 (Sum of two idempotents). Let $\mathbf{P}, \mathbf{Q} \in \mathbb{R}^{n \times n}$ be idempotent. Then

$$\mathbf{P} + \mathbf{Q} \text{ is idempotent} \iff \mathbf{PQ} = \mathbf{QP} = \mathbf{O}.$$

Under these equivalent conditions,

$$\text{col}(\mathbf{P} + \mathbf{Q}) = \text{col}(\mathbf{P}) \oplus \text{col}(\mathbf{Q}), \quad \text{rank}(\mathbf{P} + \mathbf{Q}) = \text{rank}(\mathbf{P}) + \text{rank}(\mathbf{Q}).$$

Origin: H Lemma 18.4.3; A&M Ex. 8.64

Proof.

$$(\mathbf{P} + \mathbf{Q})^2 - (\mathbf{P} + \mathbf{Q}) = \mathbf{PQ} + \mathbf{QP}.$$

If the left side is zero, premultiplication and postmultiplication by $\mathbf{P}$ give

$$\mathbf{PQ} + \mathbf{PQP} = \mathbf{O}, \quad \mathbf{PQP} + \mathbf{QP} = \mathbf{O}.$$

Hence $\mathbf{PQ} = \mathbf{QP}$; together with $\mathbf{PQ} + \mathbf{QP} = \mathbf{O}$, this gives $\mathbf{PQ} = \mathbf{QP} = \mathbf{O}$. The converse follows

from the first display.

If $\mathbf{z} \in \text{col}(\mathbf{P}) \cap \text{col}(\mathbf{Q})$, then $\mathbf{z} = \mathbf{P}\mathbf{x} = \mathbf{Q}\mathbf{y}$ and

$$\mathbf{z} = \mathbf{P}\mathbf{z} = \mathbf{P}\mathbf{Q}\mathbf{y} = \mathbf{0}.$$

The inclusion $\text{col}(\mathbf{P} + \mathbf{Q}) \subseteq \text{col}(\mathbf{P}) + \text{col}(\mathbf{Q})$ is immediate. Conversely,

$$\mathbf{P}\mathbf{x} + \mathbf{Q}\mathbf{y} = (\mathbf{P} + \mathbf{Q})(\mathbf{P}\mathbf{x} + \mathbf{Q}\mathbf{y}).$$

Thus the ranges are equal and their dimensions add. □

Proposition 10.11 (Commuting product). If idempotents $\mathbf{P}$ and $\mathbf{Q}$ commute, then $\mathbf{P}\mathbf{Q}$ is idempotent and

$$\text{col}(\mathbf{P}\mathbf{Q}) = \text{col}(\mathbf{P}) \cap \text{col}(\mathbf{Q}).$$

If $\mathbf{P}$ and $\mathbf{Q}$ are also symmetric, then $\mathbf{P}\mathbf{Q}$ is the orthogonal projector onto this intersection.

Origin: Course formulation

Proof. Commutation gives

$$(\mathbf{PQ})^2 = \mathbf{PQPQ} = \mathbf{P}^2\mathbf{Q}^2 = \mathbf{PQ}.$$

If $\mathbf{z} = \mathbf{PQx}$, then $\mathbf{Pz} = \mathbf{z} = \mathbf{Qz}$, so $\mathbf{z} \in \text{col}(\mathbf{P}) \cap \text{col}(\mathbf{Q})$. If $\mathbf{z} \in \text{col}(\mathbf{P}) \cap \text{col}(\mathbf{Q})$, then $\mathbf{PQz} = \mathbf{Pz} = \mathbf{z}$. Symmetry follows from $(\mathbf{PQ})' = \mathbf{QP} = \mathbf{PQ}$; apply Theorem 10.7. $\square$

Example 10.2 (An idempotent product without commutation). For $c \neq 0$,

$$\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} 1 & 0 \\ c & 0 \end{pmatrix}$$

are idempotent, while

$$\mathbf{PQ} = \mathbf{P}, \quad \mathbf{QP} = \mathbf{Q} \neq \mathbf{P}.$$

Thus idempotency of $\mathbf{PQ}$ does not imply commutation for general idempotents.

Origin: Course formulation

2 Quadratic forms and inertia

2.1 Polarization, entrywise products, and inertia

Definition 10.12 (Bilinear form). For $\mathbf{A} \in \mathbb{R}^{n \times n}$, define

$$b_{\mathbf{A}}(\mathbf{x}, \mathbf{y}) = \mathbf{x}' \mathbf{A} \mathbf{y}.$$

For scalars α, β ,

$$b_{\mathbf{A}}(\alpha \mathbf{x}_1 + \beta \mathbf{x}_2, \mathbf{y}) = \alpha b_{\mathbf{A}}(\mathbf{x}_1, \mathbf{y}) + \beta b_{\mathbf{A}}(\mathbf{x}_2, \mathbf{y}),$$

$$b_{\mathbf{A}}(\mathbf{x}, \alpha \mathbf{y}_1 + \beta \mathbf{y}_2) = \alpha b_{\mathbf{A}}(\mathbf{x}, \mathbf{y}_1) + \beta b_{\mathbf{A}}(\mathbf{x}, \mathbf{y}_2).$$

Thus $b_{\mathbf{A}}$ is a **bilinear form**; it is symmetric when $\mathbf{A}' = \mathbf{A}$.

Origin: H §14.1, pp. 265–266; S §1.3

Theorem 10.13 (Polarization identity). If $\mathbf{A}' = \mathbf{A}$, then

$$b_{\mathbf{A}}(\mathbf{x}, \mathbf{y}) = \frac{1}{2}\{q_{\mathbf{A}}(\mathbf{x} + \mathbf{y}) - q_{\mathbf{A}}(\mathbf{x}) - q_{\mathbf{A}}(\mathbf{y})\} = \frac{1}{4}\{q_{\mathbf{A}}(\mathbf{x} + \mathbf{y}) - q_{\mathbf{A}}(\mathbf{x} - \mathbf{y})\}.$$

Consequently, $q_{\mathbf{A}}$ determines $b_{\mathbf{A}}$ and $\mathbf{A}$.

Origin: H Ex. 14.1; A&M Ex. 8.3

Proof.

$$\begin{aligned} q_{\mathbf{A}}(\mathbf{x} + \mathbf{y}) - q_{\mathbf{A}}(\mathbf{x}) - q_{\mathbf{A}}(\mathbf{y}) &= \mathbf{x}'\mathbf{A}\mathbf{y} + \mathbf{y}'\mathbf{A}\mathbf{x} = 2\mathbf{x}'\mathbf{A}\mathbf{y}, \\ q_{\mathbf{A}}(\mathbf{x} + \mathbf{y}) - q_{\mathbf{A}}(\mathbf{x} - \mathbf{y}) &= 2\mathbf{x}'\mathbf{A}\mathbf{y} + 2\mathbf{y}'\mathbf{A}\mathbf{x} \\ &= 4\mathbf{x}'\mathbf{A}\mathbf{y}. \end{aligned}$$

Finally,

$$a_{ij} = b_{\mathbf{A}}(\mathbf{e}_i, \mathbf{e}_j).$$

□

Definition 10.14 (Hadamard product). For matrices $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$ of the same size, their **Hadamard product**, or **entrywise product**, is

$$\mathbf{A} \circ \mathbf{B} = (a_{ij}b_{ij}).$$

Origin: H Ex. 14.31; A&M Ex. 12.32; S Ex. 6.9

Definition 10.15 (Inertia). If symmetric $\mathbf{A} \in \mathbb{R}^{n \times n}$ has $n_+(\mathbf{A})$ positive, $n_-(\mathbf{A})$ negative, and $n_0(\mathbf{A})$ zero eigenvalues, counted with multiplicity, its **inertia** is

$$\text{In}(\mathbf{A}) = (n_+(\mathbf{A}), n_-(\mathbf{A}), n_0(\mathbf{A})).$$

For the 0×0 matrix, $\text{In}(\mathbf{A}) = (0, 0, 0)$.

Origin: H Ex. 14.11(c); S §5.4

2.2 Canonical signs and Sylvester's law

Theorem 10.16 (Canonical form under congruence). For every symmetric $\mathbf{A} \in \mathbb{R}^{n \times n}$ there is a nonsingular $\mathbf{T}$ such that

$$\mathbf{T}'\mathbf{A}\mathbf{T} = \text{diag}(\mathbf{I}_p, -\mathbf{I}_q, \mathbf{O}_s), \quad p + q + s = n,$$

where

$$(p, q, s) = \text{In}(\mathbf{A}).$$

Origin: H Cor. 14.3.5 and Ex. 14.11(b); S §5.4

Proof. By the symmetric spectral theorem,

$$\mathbf{A} = \mathbf{Q} \text{diag}(\lambda_1, \dots, \lambda_n) \mathbf{Q}', \quad \mathbf{Q}'\mathbf{Q} = \mathbf{I}_n.$$

After permuting the diagonal, define

$$d_i = \begin{cases} |\lambda_i|^{-1/2}, & \lambda_i \neq 0, \\ 1, & \lambda_i = 0, \end{cases} \quad \mathbf{T} = \mathbf{Q} \text{diag}(d_1, \dots, d_n).$$

Then $\mathbf{T}$ is nonsingular and

$$\mathbf{T}'\mathbf{A}\mathbf{T} = \text{diag}(d_1^2\lambda_1, \dots, d_n^2\lambda_n) = \text{diag}(\mathbf{I}_p, -\mathbf{I}_q, \mathbf{O}_s).$$

The $n = 0$ case is the empty identity. □

Lemma 10.17 (Maximal definite subspaces). For symmetric $\mathbf{A}$,

$$n_+(\mathbf{A}) = \max\{\dim \mathcal{S} : q_{\mathbf{A}}(\mathbf{x}) > 0 \text{ for every } \mathbf{0} \neq \mathbf{x} \in \mathcal{S}\},$$

$$n_-(\mathbf{A}) = \max\{\dim \mathcal{S} : q_{\mathbf{A}}(\mathbf{x}) < 0 \text{ for every } \mathbf{0} \neq \mathbf{x} \in \mathcal{S}\}.$$

For $n_+(\mathbf{A}) = 0$ or $n_-(\mathbf{A}) = 0$, the maximum is attained by $\{\mathbf{0}\}$.

Origin: Adapted from H Ex. 14.11(a); S §5.4

Proof. Let $\mathcal{E}_+, \mathcal{E}_-, \mathcal{E}_0$ be the mutually orthogonal spans of eigenvectors with positive, negative, and zero eigenvalues. The form is positive definite on $\mathcal{E}_+$, so the first maximum is at least $\dim \mathcal{E}_+ = n_+(\mathbf{A})$. If $\dim \mathcal{S} > n_+(\mathbf{A})$, then

$$\dim \mathcal{S} + \dim(\mathcal{E}_- \oplus \mathcal{E}_0) > n,$$

so $\mathcal{S} \cap (\mathcal{E}_- \oplus \mathcal{E}_0)$ contains a nonzero $\mathbf{x}$. For this vector, $q_{\mathbf{A}}(\mathbf{x}) \leq 0$. Thus the first maximum is $n_+(\mathbf{A})$. Apply the same argument to $-\mathbf{A}$ for the second formula. $\square$

Theorem 10.18 (Sylvester's law of inertia). Congruent real symmetric matrices have the same inertia. Equivalently,

$$\text{In}(\mathbf{T}'\mathbf{A}\mathbf{T}) = \text{In}(\mathbf{A})$$

for every symmetric $\mathbf{A}$ and every nonsingular $\mathbf{T}$.

Origin: H Ex. 14.11(c); S §5.4

Proof. Let $\mathbf{B} = \mathbf{T}'\mathbf{A}\mathbf{T}$. The bijection $\mathcal{S} \mapsto \mathbf{T}\mathcal{S}$ preserves dimension, and

$$q_{\mathbf{B}}(\mathbf{y}) = q_{\mathbf{A}}(\mathbf{T}\mathbf{y}).$$

Lemma 10.17 gives $n_+(\mathbf{B}) = n_+(\mathbf{A})$ and $n_-(\mathbf{B}) = n_-(\mathbf{A})$. The nonsingularity of $\mathbf{T}$ gives

$$\text{rank}(\mathbf{B}) = \text{rank}(\mathbf{T}'\mathbf{A}\mathbf{T}) = \text{rank}(\mathbf{A}), \quad n_0(\mathbf{B}) = n - \text{rank}(\mathbf{B}) = n_0(\mathbf{A}).$$

$\square$

2.3 Completing the square

Theorem 10.19 (Block completion and inertia additivity). Let $k \geq 1$, $m \geq 0$, and

$$\mathbf{A} = \begin{pmatrix} \mathbf{B} & \mathbf{C} \\ \mathbf{C}' & \mathbf{D} \end{pmatrix}, \quad \mathbf{B}' = \mathbf{B} \in \mathbb{R}^{k \times k}, \quad \mathbf{D}' = \mathbf{D} \in \mathbb{R}^{m \times m},$$

where $\mathbf{B}$ is nonsingular. With the symmetric matrix $\mathbf{S} = \mathbf{D} - \mathbf{C}'\mathbf{B}^{-1}\mathbf{C}$,

$$\begin{pmatrix} \mathbf{x} \\ \mathbf{z} \end{pmatrix}' \mathbf{A} \begin{pmatrix} \mathbf{x} \\ \mathbf{z} \end{pmatrix} = (\mathbf{x} + \mathbf{B}^{-1}\mathbf{C}\mathbf{z})' \mathbf{B} (\mathbf{x} + \mathbf{B}^{-1}\mathbf{C}\mathbf{z}) + \mathbf{z}' \mathbf{S} \mathbf{z},$$

$$\text{In}(\mathbf{A}) = \text{In}(\mathbf{B}) + \text{In}(\mathbf{S}),$$

where triples are added componentwise. If $m = 0$, $\mathbf{S}$ is the 0×0 matrix and the formula reduces to $\text{In}(\mathbf{A}) = \text{In}(\mathbf{B})$.

Origin: H Thm. 14.8.4; A&M Ex. 8.45

Proof. Since $(\mathbf{B}^{-1})' = (\mathbf{B}')^{-1} = \mathbf{B}^{-1}$, $\mathbf{S}' = \mathbf{S}$. Set

$$\mathbf{T} = \begin{pmatrix} \mathbf{I}_k & -\mathbf{B}^{-1}\mathbf{C} \\ \mathbf{0} & \mathbf{I}_m \end{pmatrix}.$$

Then $\mathbf{T}$ is nonsingular and

$$\mathbf{T}'\mathbf{A}\mathbf{T} = \begin{pmatrix} \mathbf{B} & \mathbf{0} \\ \mathbf{0} & \mathbf{S} \end{pmatrix}.$$

Substitution of $\mathbf{x} = (\mathbf{x} + \mathbf{B}^{-1}\mathbf{C}\mathbf{z}) - \mathbf{B}^{-1}\mathbf{C}\mathbf{z}$ gives the quadratic identity. The eigenvalues of a block-diagonal matrix are those of its diagonal blocks. Apply Theorem 10.18. $\square$

Corollary 10.20 (Affine completion). Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be positive definite, $\mathbf{b} \in \mathbb{R}^n$, and $c \in \mathbb{R}$. Then

$$\mathbf{x}'\mathbf{A}\mathbf{x} - 2\mathbf{b}'\mathbf{x} + c = (\mathbf{x} - \mathbf{A}^{-1}\mathbf{b})'\mathbf{A}(\mathbf{x} - \mathbf{A}^{-1}\mathbf{b}) + c - \mathbf{b}'\mathbf{A}^{-1}\mathbf{b}.$$

Hence the unique minimizer is $\mathbf{A}^{-1}\mathbf{b}$, and the minimum is $c - \mathbf{b}'\mathbf{A}^{-1}\mathbf{b}$.

Origin: A&M Ex. 8.15; H Thm. 14.12.20

Proof. Expansion gives the identity. Positive definiteness gives

$$(\mathbf{x} - \mathbf{A}^{-1}\mathbf{b})' \mathbf{A} (\mathbf{x} - \mathbf{A}^{-1}\mathbf{b}) \geq 0,$$

with equality if and only if $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$. □

Example 10.3 (Inertia by completion). For

$$\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix},$$

$$(x \ y) \mathbf{A} \begin{pmatrix} x \\ y \end{pmatrix} = 2 \left(x + \frac{y}{2} \right)^2 - \frac{5}{2} y^2.$$

Thus $\text{In}(\mathbf{A}) = (1, 1, 0)$ without solving its characteristic equation.

Origin: Adapted from A&M Ex. 8.2(c)

3 Orthogonal sums and second moments

3.1 The algebraic Cochran criterion

Definition 10.21 (Orthogonal resolution of the identity). A finite family of symmetric matrices $\mathbf{P}_1, \dots, \mathbf{P}_k \in \mathbb{R}^{n \times n}$ is an **orthogonal resolution of the identity** if

$$\sum_{i=1}^k \mathbf{P}_i = \mathbf{I}_n, \quad \mathbf{P}_i^2 = \mathbf{P}_i, \quad \mathbf{P}_i \mathbf{P}_j = \mathbf{O} \quad (i \neq j).$$

For $n = 0$, the empty family is a resolution; for $n > 0$, a resolution has at least one member.

Origin: Adapted from H Thm. 18.4.5; A&M Ex. 8.72

Theorem 10.22 (Algebraic Cochran rank criterion). Let $k \geq 1$ and let symmetric $\mathbf{A}_1, \dots, \mathbf{A}_k \in \mathbb{R}^{n \times n}$ satisfy

$$\sum_{i=1}^k \mathbf{A}_i = \mathbf{I}_n.$$

The following are equivalent:

- (i) $\mathbf{A}_i^2 = \mathbf{A}_i$ for every i ;

(ii) $\mathbf{A}_i \mathbf{A}_j = \mathbf{O}$ for every $i \neq j$;

(iii) $\sum_{i=1}^k \text{rank}(\mathbf{A}_i) = n$.

Under these conditions,

$$\mathbb{R}^n = \text{col}(\mathbf{A}_1) \oplus \cdots \oplus \text{col}(\mathbf{A}_k)$$

is an orthogonal direct sum.

Origin: H Thm. 18.4.5; A&M Ex. 8.72

Proof. (ii) $\Rightarrow$ (i):

$$\mathbf{A}_i = \mathbf{A}_i \mathbf{I}_n = \sum_{j=1}^k \mathbf{A}_i \mathbf{A}_j = \mathbf{A}_i^2.$$

(i) $\Rightarrow$ (iii), by Corollary 10.6:

$$\sum_{i=1}^k \text{rank}(\mathbf{A}_i) = \sum_{i=1}^k \text{tr}(\mathbf{A}_i) = \text{tr} \left(\sum_{i=1}^k \mathbf{A}_i \right) = n.$$

Assume (iii). Since every $\mathbf{x}$ satisfies $\mathbf{x} = \sum_i \mathbf{A}_i \mathbf{x}$,

$$\mathbb{R}^n = \sum_{i=1}^k \text{col}(\mathbf{A}_i).$$

The dimension of this sum is $n = \sum_i \dim \text{col}(\mathbf{A}_i)$, so the sum is direct. For $\mathbf{y} = \mathbf{A}_i \mathbf{x} \in \text{col}(\mathbf{A}_i)$,

$$\mathbf{y} = \sum_{j=1}^k \mathbf{A}_j \mathbf{y}$$

is its unique decomposition. Hence $\mathbf{A}_i \mathbf{y} = \mathbf{y}$ and $\mathbf{A}_j \mathbf{y} = \mathbf{0}$ for $j \neq i$, which gives $\mathbf{A}_i^2 = \mathbf{A}_i$ and $\mathbf{A}_j \mathbf{A}_i = \mathbf{0}$. Thus (iii) $\Rightarrow$ (i),(ii). Finally, symmetry gives

$$(\mathbf{A}_i \mathbf{x})' (\mathbf{A}_j \mathbf{y}) = \mathbf{x}' \mathbf{A}_i \mathbf{A}_j \mathbf{y} = 0 \quad (i \neq j),$$

so the direct sum is orthogonal. The proof includes $n = 0$. □

Corollary 10.23 (Quadratic decomposition). For an orthogonal resolution $\mathbf{P}_1, \dots, \mathbf{P}_k$ and every $\mathbf{y} \in \mathbb{R}^n$,

$$\mathbf{y} = \sum_{i=1}^k \mathbf{P}_i \mathbf{y}, \quad \|\mathbf{y}\|^2 = \sum_{i=1}^k \|\mathbf{P}_i \mathbf{y}\|^2 = \sum_{i=1}^k \mathbf{y}' \mathbf{P}_i \mathbf{y}, \quad \sum_{i=1}^k \text{rank}(\mathbf{P}_i) = n.$$

Origin: H Thm. 18.4.4; A&M Ex. 8.72

Proof. The first identity uses $\sum_i \mathbf{P}_i = \mathbf{I}_n$. Pairwise orthogonality and symmetry give

$$\|\mathbf{y}\|^2 = \left\| \sum_i \mathbf{P}_i \mathbf{y} \right\|^2 = \sum_i \|\mathbf{P}_i \mathbf{y}\|^2, \quad \|\mathbf{P}_i \mathbf{y}\|^2 = \mathbf{y}' \mathbf{P}_i' \mathbf{P}_i \mathbf{y} = \mathbf{y}' \mathbf{P}_i \mathbf{y}.$$

The rank identity is Theorem [10.22](#). □

3.2 Regression projectors and sums of squares

Theorem 10.24 (Three regression projectors). Let $1 \leq p \leq n$, let $\mathbf{X} \in \mathbb{R}^{n \times p}$ have rank p , and assume $\mathbf{1} \in \text{col}(\mathbf{X})$. Define

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \quad \mathbf{H}_0 = \frac{1}{n}\mathbf{1}\mathbf{1}', \quad \mathbf{P} = \mathbf{H} - \mathbf{H}_0, \quad \mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

Then $\mathbf{H}_0, \mathbf{P}, \mathbf{M}$ form an orthogonal resolution of $\mathbf{I}_n$, with

$$\text{rank}(\mathbf{H}_0) = 1, \quad \text{rank}(\mathbf{P}) = p - 1, \quad \text{rank}(\mathbf{M}) = n - p.$$

Thus $p = 1$ gives $\mathbf{P} = \mathbf{O}$, and $p = n$ gives $\mathbf{M} = \mathbf{O}$.

Origin: CLM Thm. 6.1; H Thm. 18.4.5

Proof. Direct multiplication gives

$$\mathbf{H}' = \mathbf{H}, \quad \mathbf{H}^2 = \mathbf{H}, \quad \text{col}(\mathbf{H}) = \text{col}(\mathbf{X}), \quad \mathbf{H}_0' = \mathbf{H}_0, \quad \mathbf{H}_0^2 = \mathbf{H}_0.$$

Since $\mathbf{1} \in \text{col}(\mathbf{X})$,

$$\mathbf{H}\mathbf{1} = \mathbf{1}, \quad \mathbf{H}\mathbf{H}_0 = \mathbf{H}_0, \quad \mathbf{H}_0\mathbf{H} = (\mathbf{H}\mathbf{H}_0)' = \mathbf{H}_0.$$

Hence

$$\mathbf{P}^2 = (\mathbf{H} - \mathbf{H}_0)^2 = \mathbf{H} - \mathbf{H}_0 = \mathbf{P},$$

$$\mathbf{M}^2 = (\mathbf{I}_n - \mathbf{H})^2 = \mathbf{M},$$

$$\mathbf{H}_0\mathbf{P} = \mathbf{O},$$

$$\mathbf{H}_0\mathbf{M} = \mathbf{O},$$

$$\mathbf{P}\mathbf{M} = \mathbf{O},$$

$$\mathbf{H}_0 + \mathbf{P} + \mathbf{M} = \mathbf{I}_n.$$

All three matrices are symmetric. Corollary 10.6 and $\text{rank}(\mathbf{H}) = \text{rank}(\mathbf{X}) = p$ give

$$\text{rank}(\mathbf{H}_0) = 1,$$

$$\text{rank}(\mathbf{P}) = \text{tr}(\mathbf{H}) - \text{tr}(\mathbf{H}_0) = p - 1,$$

$$\text{rank}(\mathbf{M}) = n - \text{rank}(\mathbf{H}) = n - p.$$

An idempotent of rank zero is the zero matrix. □

Corollary 10.25 (Total, fitted, and residual sums of squares). Under the hypotheses of Theorem 10.24, for every $\mathbf{y} \in \mathbb{R}^n$ define

$$\text{TSS} = \mathbf{y}'(\mathbf{I}_n - \mathbf{H}_0)\mathbf{y}, \quad \text{SSR} = \mathbf{y}'\mathbf{P}\mathbf{y}, \quad \text{SSE} = \mathbf{y}'\mathbf{M}\mathbf{y}.$$

Then

$$\text{TSS} = \text{SSR} + \text{SSE}, \quad \text{TSS}, \text{SSR}, \text{SSE} \geq 0.$$

Origin: CLM Thm. 6.1

Proof. Theorem 10.24 gives $\mathbf{I}_n - \mathbf{H}_0 = \mathbf{P} + \mathbf{M}$. Hence

$$\text{TSS} = \mathbf{y}'(\mathbf{P} + \mathbf{M})\mathbf{y} = \text{SSR} + \text{SSE}.$$

Each matrix is symmetric idempotent, hence positive semidefinite by Corollary 10.9. □

Example 10.4 (A degenerate fitted component). If $p = 1$ and $\mathbf{X} = \mathbf{1}$, then

$$\mathbf{H} = \mathbf{H}_0, \quad \mathbf{P} = \mathbf{O}, \quad \text{SSR} = 0, \quad \text{TSS} = \text{SSE}.$$

If $p = n$, then $\mathbf{H} = \mathbf{I}_n$, $\mathbf{M} = \mathbf{O}$, and $\text{SSE} = 0$ for every $\mathbf{y}$.

Origin: CLM Thm. 6.1

3.3 Second-moment identities

Theorem 10.26 (Expectation of a quadratic form). Let $\mathbf{y} = (Y_1, \dots, Y_n)'$ have finite second moments, mean $\boldsymbol{\mu}$, and covariance matrix $\boldsymbol{\Sigma} = (\sigma_{ij})$. For every constant $\mathbf{A} \in \mathbb{R}^{n \times n}$,

$$\mathbb{E}[\mathbf{y}'\mathbf{A}\mathbf{y}] = \text{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}.$$

The formula also holds for nonsymmetric $\mathbf{A}$.

Origin: CLM Lemma 3.2

Proof. For $\mu_i = \mathbb{E}[Y_i]$,

$$\sigma_{ij} = \mathbb{E}[(Y_i - \mu_i)(Y_j - \mu_j)] = \mathbb{E}[Y_i Y_j] - \mu_i \mu_j.$$

Hence, using $\sigma_{ij} = \sigma_{ji}$,

$$\begin{aligned}\mathbf{E}[\mathbf{y}'\mathbf{A}\mathbf{y}] &= \sum_{i=1}^n \sum_{j=1}^n a_{ij} \mathbf{E}[Y_i Y_j] \\ &= \sum_{i,j} a_{ij} \sigma_{ij} + \sum_{i,j} a_{ij} \mu_i \mu_j \\ &= \sum_i \sum_j a_{ij} \sigma_{ji} + \boldsymbol{\mu}' \mathbf{A} \boldsymbol{\mu} \\ &= \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}' \mathbf{A} \boldsymbol{\mu}.\end{aligned}$$

□

Corollary 10.27 (Expected orthogonal components). If $\mathbf{P}_1, \dots, \mathbf{P}_k$ is an orthogonal resolution of $\mathbf{I}_n$ and

$$\mathbf{E}[\mathbf{y}] = \boldsymbol{\mu}, \quad \boldsymbol{\Sigma} = \sigma^2 \mathbf{I}_n, \quad \sigma^2 \geq 0,$$

then

$$\mathbb{E}[\mathbf{y}'\mathbf{P}_i\mathbf{y}] = \sigma^2 \text{rank}(\mathbf{P}_i) + \boldsymbol{\mu}'\mathbf{P}_i\boldsymbol{\mu}, \quad i = 1, \dots, k.$$

Summing these identities gives

$$\mathbb{E}[\mathbf{y}'\mathbf{y}] = n\sigma^2 + \boldsymbol{\mu}'\boldsymbol{\mu}.$$

Origin: H Cor. 10.2.2; CLM Lemma 3.2

Proof. Theorem 10.26 and Corollary 10.6 give

$$\mathbb{E}[\mathbf{y}'\mathbf{P}_i\mathbf{y}] = \sigma^2 \text{tr}(\mathbf{P}_i) + \boldsymbol{\mu}'\mathbf{P}_i\boldsymbol{\mu} = \sigma^2 \text{rank}(\mathbf{P}_i) + \boldsymbol{\mu}'\mathbf{P}_i\boldsymbol{\mu}.$$

Use $\sum_i \mathbf{P}_i = \mathbf{I}_n$ and $\sum_i \text{rank}(\mathbf{P}_i) = n$ to sum. □

Theorem 10.28 (Expectation of the residual mean square). Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ have rank $p < n$, and suppose

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}, \quad \mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'] = \sigma^2\mathbf{I}_n.$$

With $\mathbf{M} = \mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, define

$$\text{SSE} = \mathbf{y}'\mathbf{M}\mathbf{y}, \quad \text{MSE} = \frac{\text{SSE}}{n - p}.$$

Then

$$\mathbf{E}[\text{SSE}] = \sigma^2(n - p), \quad \mathbf{E}[\text{MSE}] = \sigma^2.$$

If instead $p = n$, $\mathbf{M} = \mathbf{O}$ and $\text{SSE} = 0$; the displayed quotient is undefined.

Origin: CLM Thm. 3.4; H Cor. 10.2.2

Proof. Since $\mathbf{M}\mathbf{X} = \mathbf{O}$,

$$\mathbf{M}\mathbf{y} = \mathbf{M}\boldsymbol{\varepsilon}, \quad \text{SSE} = \mathbf{y}'\mathbf{M}'\mathbf{M}\mathbf{y} = \boldsymbol{\varepsilon}'\mathbf{M}\boldsymbol{\varepsilon}.$$

Theorem 10.26 and Corollary 10.6 give

$$\mathbf{E}[\text{SSE}] = \text{tr}(\mathbf{M}\sigma^2\mathbf{I}_n) = \sigma^2 \text{rank}(\mathbf{M}) = \sigma^2(n - p).$$

Division by $n - p > 0$ proves the second identity. If $p = n$, the full-rank square $\mathbf{X}$ has $\text{col}(\mathbf{X}) = \mathbb{R}^n$, so $\mathbf{H} = \mathbf{I}_n$ and $\mathbf{M} = \mathbf{O}$. □

4 Computation: Numerical Checks in R

4.1 An oblique projector and its canonical form

$$\mathbf{P} = \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{T} = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}.$$

```

1 P <- matrix(c(1, 0, 2, 0), 2, 2)
2 Tmat <- matrix(c(1, 0, -2, 1), 2, 2)
3 D <- diag(c(1, 0))
4 c(idempotency_residual = max(abs(P %*% P - P)),
5   canonical_residual = max(abs(solve(Tmat) %*% P %*% Tmat - D)),
6   trace = sum(diag(P)),
7   rank_qr_tol_1e_12 = qr(P, tol = 1e-12)$rank)

```

```

## idempotency_residual    canonical_residual
##                      0                      0
##                trace    rank_qr_tol_1e_12
##                      1                      1

```

The columns of $\mathbf{T}$ are bases of $\text{col}(\mathbf{P})$ and $\text{null}(\mathbf{P})$. The rank is computed by QR with tolerance 10^{-12} .

4.2 Commuting and noncommuting products

```

1 P <- diag(c(1, 0, 0))
2 Q_comm <- diag(c(1, 1, 0))
3 Q_noncomm <- matrix(c(1, 1, 0,
4                       0, 0, 0,
5                       0, 0, 1), 3, 3, byrow = TRUE)
6 prod_comm <- P %*% Q_comm
7 prod_noncomm <- P %*% Q_noncomm
8 c(commutator_comm = max(abs(P %*% Q_comm - Q_comm %*% P)),
9   product_comm_idempotency = max(abs(prod_comm %*% prod_comm - prod_comm)),
10  factor_noncomm_idempotency = max(abs(
11    Q_noncomm %*% Q_noncomm - Q_noncomm)),
12  commutator_noncomm = max(abs(P %*% Q_noncomm - Q_noncomm %*% P)),
13  product_noncomm_idempotency = max(abs(
14    prod_noncomm %*% prod_noncomm - prod_noncomm)))

```

```

##          commutator_comm    product_comm_idempotency
##                0                0
## factor_noncomm_idempotency    commutator_noncomm
##                0                1

```

```
## product_noncomm_idempotency
##                                0
```

The second product happens to be idempotent although its factors do not commute, as in Example 10.2.

4.3 Completion against a congruence transform

```
1 A <- matrix(c(2, 1, 1, -2), 2, 2)
2 Tmat <- matrix(c(1, 0, -1/2, 1), 2, 2)
3 D <- diag(c(2, -5/2))
4 lam <- eigen(A, symmetric = TRUE)$values
5 c(congruence_residual = max(abs(t(Tmat) %*% A %*% Tmat - D)),
6   positive_eigenvalues = sum(lam > 1e-12),
7   negative_eigenvalues = sum(lam < -1e-12),
8   zero_eigenvalues = sum(abs(lam) <= 1e-12))
```

```
## congruence_residual positive_eigenvalues
##                   0                   1
## negative_eigenvalues zero_eigenvalues
##                   1                   0
```

Eigenvalue signs use the stated tolerance 10^{-12} ; the manual congruence gives the same inertia.

4.4 Exact finite-support second moments

```

1 support <- rbind(c(0, 0), c(2, 0), c(0, 2), c(2, 2))
2 prob <- c(1, 2, 1, 4) / 8
3 A <- matrix(c(2, -1, -1, 3), 2, 2)
4 mu <- colSums(support * prob)
5 centered <- sweep(support, 2, mu)
6 Sigma <- t(centered) %*% sweep(centered, 1, prob, "*")
7 q_values <- rowSums((support %*% A) * support)
8 direct <- sum(prob * q_values)
9 formula <- sum(diag(A %*% Sigma)) + drop(t(mu) %*% A %*% mu)
10 c(direct_enumeration = direct,
11   trace_formula = formula,
12   difference = direct - formula)

```

##	direct_enumeration	trace_formula	difference
##	9.5	9.5	0.0

All support coordinates and probabilities are dyadic, so the printed zero does not depend on Monte Carlo sampling.

4.5 Regression resolution and sums of squares

```
1 x <- -2:2
2 X <- cbind(1, x)
3 y <- c(2, 1, 3, 6, 8)
4 n <- nrow(X)
5 H <- X %*% solve(crossprod(X), t(X))
6 H0 <- matrix(1 / n, n, n)
7 P <- H - H0
8 M <- diag(n) - H
9 pieces <- list(H0 = H0, P = P, M = M)
10 pair_residual <- max(abs(P %*% M), abs(P %*% H0), abs(M %*% H0))
11 ranks <- vapply(pieces, function(B) qr(B, tol = 1e-10)$rank, numeric(1))
12 ss <- c(TSS = drop(t(y) %*% (diag(n) - H0) %*% y),
13         SSR = drop(t(y) %*% P %*% y),
14         SSE = drop(t(y) %*% M %*% y))
15 c(identity_residual = max(abs(H0 + P + M - diag(n))),
16     pairwise_product_residual = pair_residual,
17     rank_sum_minus_n = sum(ranks) - n,
18     TSS_minus_SSR_minus_SSE = ss[1] - ss[2] - ss[3])
```

```
##          identity_residual    pairwise_product_residual
##          0.000e+00          5.995e-17
##          rank_sum_minus_n    TSS_minus_SSR_minus_SSE.TSS
##          0.000e+00          7.105e-15
```

Numerical ranks use QR with tolerance 10^{-10} .

5 Pitfalls

5.1 Trace equal to rank does not imply idempotency

$$\mathbf{A} = \text{diag}(2, 2, -1), \quad \text{tr}(\mathbf{A}) = 3 = \text{rank}(\mathbf{A}), \quad \mathbf{A}^2 = \text{diag}(4, 4, 1) \neq \mathbf{A}.$$

5.2 Eigenvalues zero and one do not suffice without diagonalizability

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (\lambda_1, \lambda_2, \lambda_3) = (0, 0, 1), \quad \mathbf{A}^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \neq \mathbf{A}.$$

5.3 An idempotent product need not have commuting factors

$$\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{PQ} = \mathbf{P}, \quad \mathbf{QP} = \mathbf{Q}.$$

5.4 Congruence does not preserve eigenvalues

$$\mathbf{A} = \mathbf{I}_2, \quad \mathbf{T} = \text{diag}(2, 1), \quad \mathbf{T}'\mathbf{A}\mathbf{T} = \text{diag}(4, 1).$$

The two spectra differ; both inertias equal $(2, 0, 0)$.

5.5 The algebraic Cochran criterion requires the sum condition

$$\mathbf{A}_1 = \mathbf{A}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

are symmetric idempotents, but $\mathbf{A}_1 + \mathbf{A}_2 \neq \mathbf{I}_2$ and $\mathbf{A}_1\mathbf{A}_2 \neq \mathbf{O}$.

5.6 The residual mean-square denominator may be zero

If $p = \text{rank}(\mathbf{X}) = n$, then $\mathbf{H} = \mathbf{I}_n$, $\mathbf{M} = \mathbf{O}$, and $\text{SSE} = 0$. The quotient $\text{SSE}/(n - p)$ is undefined.

5.7 The moment identity has no independence premise

$$E[\mathbf{y}'\mathbf{A}\mathbf{y}] = \text{tr}(\mathbf{A}\Sigma) + \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}$$

requires finite second moments. Independence, normality, and a density are absent from its hypotheses.

6 Exercises

Exercise 10.1 (Fixed points characterize the range). Let $\mathbf{P}$ be idempotent. Prove

$$\text{col}(\mathbf{P}) = \{\mathbf{x} : \mathbf{P}\mathbf{x} = \mathbf{x}\}.$$

Deduce that two idempotents with the same range and the same null space are equal.

Origin: H Thm. 11.7.1

Solution. If $\mathbf{x} = \mathbf{P}\mathbf{y}$, then $\mathbf{P}\mathbf{x} = \mathbf{P}^2\mathbf{y} = \mathbf{x}$. Conversely, $\mathbf{P}\mathbf{x} = \mathbf{x}$ implies $\mathbf{x} \in \text{col}(\mathbf{P})$.

Let $\mathbf{P}, \mathbf{Q}$ have common range $\mathcal{S}$ and common null space $\mathcal{T}$. For the unique decomposition $\mathbf{x} = \mathbf{s} + \mathbf{t}$ with $\mathbf{s} \in \mathcal{S}, \mathbf{t} \in \mathcal{T}$,

$$\mathbf{P}\mathbf{x} = \mathbf{s} = \mathbf{Q}\mathbf{x}.$$

Thus $\mathbf{P} = \mathbf{Q}$. 

Exercise 10.2 (Constructing an oblique projector). Let $\mathbf{S} \in \mathbb{R}^{n \times r}$ and $\mathbf{T} \in \mathbb{R}^{n \times (n-r)}$ have full column rank, and suppose $\mathbf{B} = (\mathbf{S}, \mathbf{T})$ is nonsingular. Show that

$$\mathbf{P} = \mathbf{B} \begin{pmatrix} \mathbf{I}_r & \mathbf{O} \\ \mathbf{O} & \mathbf{O} \end{pmatrix} \mathbf{B}^{-1}$$

is the unique projection onto $\text{col}(\mathbf{S})$ along $\text{col}(\mathbf{T})$. Include $r = 0$ and $r = n$.

Origin: Adapted from A&M Ex. 8.60; H §11.7, p. 212

Solution. Let $\mathbf{D} = \text{diag}(\mathbf{I}_r, \mathbf{O})$. Then

$$\mathbf{P}^2 = \mathbf{B}\mathbf{D}^2\mathbf{B}^{-1} = \mathbf{P}, \quad \mathbf{P}\mathbf{S} = \mathbf{S}, \quad \mathbf{P}\mathbf{T} = \mathbf{O}.$$

Hence $\text{col}(\mathbf{P}) = \text{col}(\mathbf{S})$ and $\text{null}(\mathbf{P}) = \text{col}(\mathbf{T})$. Uniqueness is Exercise 10.1. For $r = 0$, $\mathbf{P} = \mathbf{O}$; for $r = n$, $\mathbf{P} = \mathbf{I}_n$. 

Exercise 10.3 (Contractive idempotents are orthogonal). Let $\mathbf{P}$ be idempotent. Prove

$$\|\mathbf{P}\mathbf{x}\| \leq \|\mathbf{x}\| \quad (\mathbf{x} \in \mathbb{R}^n) \iff \mathbf{P}' = \mathbf{P}.$$

Deduce that every nonsymmetric idempotent expands at least one vector.

Origin: Course formulation from H Thm. 12.3.9 and §12.1

Solution. If $\mathbf{P}' = \mathbf{P}$, then

$$\mathbf{x} = \mathbf{P}\mathbf{x} + (\mathbf{I}_n - \mathbf{P})\mathbf{x}, \quad (\mathbf{P}\mathbf{x})'(\mathbf{I}_n - \mathbf{P})\mathbf{x} = 0,$$

so

$$\|\mathbf{x}\|^2 = \|\mathbf{P}\mathbf{x}\|^2 + \|(\mathbf{I}_n - \mathbf{P})\mathbf{x}\|^2 \geq \|\mathbf{P}\mathbf{x}\|^2.$$

Conversely, let $\mathbf{s} \in \text{col}(\mathbf{P})$ and $\mathbf{t} \in \text{null}(\mathbf{P})$. For every $\alpha \in \mathbb{R}$,

$$\|\mathbf{s}\|^2 = \|\mathbf{P}(\mathbf{s} + \alpha\mathbf{t})\|^2 \leq \|\mathbf{s} + \alpha\mathbf{t}\|^2 = \|\mathbf{s}\|^2 + 2\alpha\mathbf{s}'\mathbf{t} + \alpha^2\|\mathbf{t}\|^2.$$

The quadratic on the right is nonnegative for every α , hence $\mathbf{s}'\mathbf{t} = 0$. Thus $\text{col}(\mathbf{P}) \perp \text{null}(\mathbf{P})$, and Theorem 10.7 gives $\mathbf{P}' = \mathbf{P}$. The contrapositive gives the last assertion. ■

Exercise 10.4 (Products of orthogonal projectors). Let $\mathbf{P}$ and $\mathbf{Q}$ be symmetric idempotents. Prove

$$\mathbf{PQ} \text{ is idempotent} \iff \mathbf{PQ} = \mathbf{QP}.$$

In that case, identify its range.

Origin: Course formulation

Solution. The reverse implication is Proposition 10.11. Suppose $(\mathbf{PQ})^2 = \mathbf{PQ}$ and set $\mathbf{C} = \mathbf{PQ} - \mathbf{QP}$. Transposition gives $(\mathbf{QP})^2 = \mathbf{QP}$. Since

$$\mathrm{tr}(\mathbf{C}'\mathbf{C}) = \mathrm{tr}(\mathbf{QPQ}) - \mathrm{tr}((\mathbf{QP})^2) - \mathrm{tr}((\mathbf{PQ})^2) + \mathrm{tr}(\mathbf{PQP}) = 0,$$

and $\mathrm{tr}(\mathbf{C}'\mathbf{C}) = \sum_{i,j} c_{ij}^2$, $\mathbf{C} = \mathbf{O}$. Thus $\mathbf{PQ} = \mathbf{QP}$. Proposition 10.11 gives

$$\mathrm{col}(\mathbf{PQ}) = \mathrm{col}(\mathbf{P}) \cap \mathrm{col}(\mathbf{Q}).$$



Exercise 10.5 (Calculus with a resolution of identity). Let $\mathbf{P}_1, \dots, \mathbf{P}_k$ be an orthogonal resolution of $\mathbf{I}_n$, and let $a_1, \dots, a_k \in \mathbb{R}$. Define $\mathbf{A} = \sum_i a_i \mathbf{P}_i$. Prove, for every polynomial f , that

$$f(\mathbf{A}) = \sum_{i=1}^k f(a_i) \mathbf{P}_i.$$

Deduce that $\mathbf{A}$ is nonsingular if and only if every a_i attached to a nonzero $\mathbf{P}_i$ is nonzero, in which case

$$\mathbf{A}^{-1} = \sum_{i: \mathbf{P}_i \neq \mathbf{0}} a_i^{-1} \mathbf{P}_i.$$

Origin: Adapted from A&M Ex. 8.73; S §5.5

Solution. For $m \geq 1$, pairwise annihilation gives

$$\mathbf{A}^m = \left(\sum_i a_i \mathbf{P}_i \right)^m = \sum_i a_i^m \mathbf{P}_i.$$

The constant term satisfies $\mathbf{I}_n = \sum_i \mathbf{P}_i$, so linearity gives the polynomial identity.

If $a_j = 0$ and $\mathbf{P}_j \neq \mathbf{O}$, choose $\mathbf{0} \neq \mathbf{x} \in \text{col}(\mathbf{P}_j)$; then $\mathbf{A}\mathbf{x} = \mathbf{0}$. If all relevant a_i are nonzero, then

$$\left(\sum_i a_i \mathbf{P}_i \right) \left(\sum_i a_i^{-1} \mathbf{P}_i \right) = \sum_i \mathbf{P}_i = \mathbf{I}_n,$$

where zero members may be omitted. 

Exercise 10.6 (Resolution of an idempotent). Let $\mathbf{P}$ be a symmetric idempotent and let symmetric matrices $\mathbf{A}_1, \dots, \mathbf{A}_k$ satisfy

$$\sum_{i=1}^k \mathbf{A}_i = \mathbf{P}.$$

Prove that the following are equivalent:

- (i) every $\mathbf{A}_i$ is idempotent;
- (ii) $\mathbf{A}_i \mathbf{A}_j = \mathbf{O}$ for $i \neq j$;
- (iii) $\sum_i \text{rank}(\mathbf{A}_i) = \text{rank}(\mathbf{P})$.

Origin: Adapted from H Thms. 18.4.1 and 18.4.5

Solution. Add $\mathbf{A}_0 = \mathbf{I}_n - \mathbf{P}$. Since

$$\mathbf{A}_0 + \sum_{i=1}^k \mathbf{A}_i = \mathbf{I}_n, \quad \text{rank}(\mathbf{A}_0) = n - \text{rank}(\mathbf{P}),$$

condition (iii) is equivalent to $\sum_{i=0}^k \text{rank}(\mathbf{A}_i) = n$. Theorem 10.22 then gives (i) and (ii). If (i) holds, Corollary 10.6 gives

$$\sum_i \text{rank}(\mathbf{A}_i) = \sum_i \text{tr}(\mathbf{A}_i) = \text{tr}(\mathbf{P}) = \text{rank}(\mathbf{P}),$$

so (iii) holds. If (ii) holds, then

$$\mathbf{0} = \mathbf{P}^2 - \mathbf{P} = \sum_i (\mathbf{A}_i^2 - \mathbf{A}_i).$$

Each $(\mathbf{A}_i^2 - \mathbf{A}_i)\mathbf{x}$ lies in $\text{col}(\mathbf{A}_i)$. These column spaces are pairwise orthogonal because the matrices are symmetric and $\mathbf{A}_i\mathbf{A}_j = \mathbf{0}$. Thus every summand is zero for every $\mathbf{x}$, so $\mathbf{A}_i^2 = \mathbf{A}_i$ and (i) holds.



Exercise 10.7 (A diagonal entry of an inverse). Let $\mathbf{A} = (a_{ij}) \succ \mathbf{O}$ and $\mathbf{A}^{-1} = (b_{ij})$. Prove, for $i = 1, \dots, n$,

$$b_{ii} \geq \frac{1}{a_{ii}},$$

with equality if and only if $a_{ij} = 0$ for every $j \neq i$.

Origin: H Ex. 14.8

Solution. Simultaneously permute rows and columns so that $i = 1$, and write

$$\mathbf{A} = \begin{pmatrix} a & \mathbf{r}' \\ \mathbf{r} & \mathbf{C} \end{pmatrix}.$$

For $n = 1$, $b_{11} = a^{-1}$. For $n > 1$, $\mathbf{C} \succ \mathbf{O}$. If the first column of $\mathbf{A}^{-1}$ is $(\beta, \mathbf{z}')'$, then

$$\mathbf{r}\beta + \mathbf{C}\mathbf{z} = \mathbf{0}, \quad a\beta + \mathbf{r}'\mathbf{z} = 1.$$

Thus

$$\mathbf{z} = -\beta\mathbf{C}^{-1}\mathbf{r}, \quad b_{11} = \beta = \frac{1}{a - \mathbf{r}'\mathbf{C}^{-1}\mathbf{r}}.$$

Block completion gives

$$a - \mathbf{r}'\mathbf{C}^{-1}\mathbf{r} > 0, \quad \mathbf{r}'\mathbf{C}^{-1}\mathbf{r} \geq 0.$$

Therefore $b_{11} \geq a^{-1}$, with equality exactly when $\mathbf{r}'\mathbf{C}^{-1}\mathbf{r} = 0$, equivalently $\mathbf{r} = \mathbf{0}$. ■

Exercise 10.8 (Two quadratic penalties). Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ be positive definite and let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$. Find the unique minimizer of

$$g(\mathbf{x}) = (\mathbf{x} - \mathbf{a})'\mathbf{A}(\mathbf{x} - \mathbf{a}) + (\mathbf{x} - \mathbf{b})'\mathbf{B}(\mathbf{x} - \mathbf{b}),$$

and give the minimum as an exact quadratic form in $\mathbf{a} - \mathbf{b}$.

Origin: Adapted from A&M Ex. 8.15

Solution. Set $\mathbf{C} = \mathbf{A} + \mathbf{B}$ and $\mathbf{d} = \mathbf{Aa} + \mathbf{Bb}$. Then

$$g(\mathbf{x}) = \mathbf{x}'\mathbf{C}\mathbf{x} - 2\mathbf{d}'\mathbf{x} + \mathbf{a}'\mathbf{Aa} + \mathbf{b}'\mathbf{Bb}.$$

Corollary 10.20 gives the minimizer

$$\mathbf{x}_* = (\mathbf{A} + \mathbf{B})^{-1}(\mathbf{Aa} + \mathbf{Bb}).$$

Let $\mathbf{u} = \mathbf{a} - \mathbf{b}$ and substitute $\mathbf{x} = \mathbf{b} + \mathbf{z}$. The minimum over $\mathbf{z}$ of

$$(\mathbf{z} - \mathbf{u})' \mathbf{A} (\mathbf{z} - \mathbf{u}) + \mathbf{z}' \mathbf{B} \mathbf{z}$$

is

$$\mathbf{u}' \{ \mathbf{A} - \mathbf{A}(\mathbf{A} + \mathbf{B})^{-1} \mathbf{A} \} \mathbf{u} = \mathbf{u}' (\mathbf{A}^{-1} + \mathbf{B}^{-1})^{-1} \mathbf{u},$$

because

$$\mathbf{A} - \mathbf{A}(\mathbf{A} + \mathbf{B})^{-1} \mathbf{A} = \mathbf{A}(\mathbf{A} + \mathbf{B})^{-1} \mathbf{B}$$

and the inverse of the last matrix is $\mathbf{B}^{-1}(\mathbf{A} + \mathbf{B})\mathbf{A}^{-1} = \mathbf{A}^{-1} + \mathbf{B}^{-1}$. 

Exercise 10.9 (A rank-one change of positive inertia). Let $n \geq 1$, $\mathbf{A} = \mathbf{A}' \in \mathbb{R}^{n \times n}$, and $\mathbf{u} \in \mathbb{R}^n$. Prove

$$n_+(\mathbf{A}) \leq n_+(\mathbf{A} + \mathbf{u}\mathbf{u}') \leq n_+(\mathbf{A}) + 1.$$

Construct examples attaining either bound.

Origin: S §5.4

Solution. If $q_{\mathbf{A}}$ is positive on $\mathcal{S} \setminus \{\mathbf{0}\}$, then

$$\mathbf{x}'(\mathbf{A} + \mathbf{u}\mathbf{u}')\mathbf{x} = \mathbf{x}'\mathbf{A}\mathbf{x} + (\mathbf{u}'\mathbf{x})^2 > 0$$

there; Lemma 10.17 gives the lower bound.

Let $\mathcal{T}$ be a maximal positive subspace for $\mathbf{A} + \mathbf{u}\mathbf{u}'$. Since $\mathcal{T} \cap \text{span}\{\mathbf{u}\}^\perp$ has dimension at least $\dim \mathcal{T} - 1$, and on this intersection

$$\mathbf{x}'(\mathbf{A} + \mathbf{u}\mathbf{u}')\mathbf{x} = \mathbf{x}'\mathbf{A}\mathbf{x},$$

Lemma 10.17 gives $n_+(\mathbf{A}) \geq \dim \mathcal{T} - 1$.

The lower bound is attained by $\mathbf{A} = \mathbf{I}_n$ and any $\mathbf{u}$. For the upper bound, for $0 \leq p < n$, take $\mathbf{A} = \text{diag}(\mathbf{I}_p, -1, \mathbf{0})$ and $\mathbf{u} = \sqrt{2}\mathbf{e}_{p+1}$; the entry -1 becomes 1. ■

Exercise 10.10 (Inertia and a Schur complement). Let

$$\mathbf{A} = \begin{pmatrix} \mathbf{B} & \mathbf{C} \\ \mathbf{C}' & \mathbf{D} \end{pmatrix}$$

be symmetric with nonsingular $\mathbf{B}$. Prove:

- (a) $\mathbf{A}$ is nonsingular if and only if $\mathbf{D} - \mathbf{C}'\mathbf{B}^{-1}\mathbf{C}$ is nonsingular;
- (b) $\mathbf{A}$ and $\mathbf{B}$ have the same positive and negative inertia counts if and only if the Schur complement is a zero matrix;
- (c) if $\mathbf{B}$ is positive definite, then $\mathbf{A}$ is positive definite if and only if its Schur complement is positive definite.

Origin: H Thm. 14.8.4; A&M Ex. 8.45

Solution. Theorem 10.19 gives

$$\text{In}(\mathbf{A}) = \text{In}(\mathbf{B}) + \text{In}(\mathbf{D} - \mathbf{C}'\mathbf{B}^{-1}\mathbf{C}).$$

- (a) A symmetric matrix is nonsingular exactly when its zero inertia count is zero. Since $\mathbf{B}$ is nonsingular, the displayed equality proves (a).

(b) The positive and negative counts contributed by the Schur complement are both zero exactly when all its eigenvalues are zero. A real symmetric matrix with only zero eigenvalues is the zero matrix.

(c) Positive definiteness is equivalent to inertia $(n, 0, 0)$; componentwise addition proves the equivalence. 

Exercise 10.11 (Nested model reduction). Let $\mathcal{S}_0 \subseteq \mathcal{S}_1 \subseteq \mathbb{R}^n$, with orthogonal projectors $\mathbf{H}_0, \mathbf{H}_1$ and residual projectors $\mathbf{M}_j = \mathbf{I}_n - \mathbf{H}_j$. Prove that

$$\mathbf{P} = \mathbf{H}_1 - \mathbf{H}_0$$

is a symmetric idempotent of rank $\dim \mathcal{S}_1 - \dim \mathcal{S}_0$, and

$$\mathbf{M}_0 = \mathbf{P} + \mathbf{M}_1, \quad \mathbf{y}'\mathbf{M}_0\mathbf{y} - \mathbf{y}'\mathbf{M}_1\mathbf{y} = \mathbf{y}'\mathbf{P}\mathbf{y} \geq 0.$$

Origin: Adapted from CLM Ex. 11.14(a); H Thm. 18.4.5

Solution. Since $\mathbf{H}_0\mathbf{x} \in \mathcal{S}_0 \subseteq \mathcal{S}_1$ for every $\mathbf{x}$,

$$\mathbf{H}_1\mathbf{H}_0\mathbf{x} = \mathbf{H}_0\mathbf{x}, \quad \mathbf{H}_0\mathbf{H}_1 = (\mathbf{H}_1\mathbf{H}_0)' = \mathbf{H}_0.$$

Hence

$$\mathbf{P}' = \mathbf{P}, \quad \mathbf{P}^2 = \mathbf{H}_1 - 2\mathbf{H}_0 + \mathbf{H}_0 = \mathbf{P}.$$

Therefore

$$\text{rank}(\mathbf{P}) = \text{tr}(\mathbf{P}) = \text{rank}(\mathbf{H}_1) - \text{rank}(\mathbf{H}_0) = \dim \mathcal{S}_1 - \dim \mathcal{S}_0.$$

Finally,

$$\mathbf{M}_0 = \mathbf{I}_n - \mathbf{H}_0 = (\mathbf{H}_1 - \mathbf{H}_0) + (\mathbf{I}_n - \mathbf{H}_1) = \mathbf{P} + \mathbf{M}_1,$$

and $\mathbf{y}'\mathbf{P}\mathbf{y} \geq 0$ by Corollary 10.9. 

Exercise 10.12 (Schur product theorem). If $\mathbf{A}, \mathbf{B} \succeq \mathbf{O}$ have the same order, prove $\mathbf{A} \circ \mathbf{B} \succeq \mathbf{O}$. If $\mathbf{A}, \mathbf{B} \succ \mathbf{O}$, prove $\mathbf{A} \circ \mathbf{B} \succ \mathbf{O}$.

Origin: H Ex. 14.31; A&M Ex. 12.32; S Ex. 6.9

Solution. For $\mathbf{D} = \text{diag}(x_1, \dots, x_n)$,

$$\mathbf{x}'(\mathbf{A} \circ \mathbf{B})\mathbf{x} = \sum_{i,j} a_{ij}b_{ij}x_i x_j = \text{tr}(\mathbf{A}\mathbf{D}\mathbf{B}\mathbf{D}).$$

Moreover, $(\mathbf{A} \circ \mathbf{B})' = \mathbf{A} \circ \mathbf{B}$. Let $\mathbf{A}^{1/2}$ and $\mathbf{B}^{1/2}$ be the symmetric square roots and set $\mathbf{M} = \mathbf{A}^{1/2}\mathbf{D}\mathbf{B}^{1/2}$. Then

$$\text{tr}(\mathbf{A}\mathbf{D}\mathbf{B}\mathbf{D}) = \text{tr}(\mathbf{M}\mathbf{M}') = \sum_{i,j} m_{ij}^2 \geq 0.$$

Thus $\mathbf{A} \circ \mathbf{B} \succeq \mathbf{O}$. If $\mathbf{A}, \mathbf{B} \succ \mathbf{O}$, their square roots are nonsingular, so

$$\mathbf{M} = \mathbf{O} \iff \mathbf{D} = \mathbf{O} \iff \mathbf{x} = \mathbf{0}.$$

Hence the quadratic form is positive for $\mathbf{x} \neq \mathbf{0}$. ■

Exercise 10.13 (A second-moment comparison). Let $\mathbf{y}$ have mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, and let symmetric $\mathbf{A}, \mathbf{B}$ satisfy $\mathbf{A} \succeq \mathbf{B}$. Prove

$$\mathbb{E}[\mathbf{y}'\mathbf{A}\mathbf{y}] \geq \mathbb{E}[\mathbf{y}'\mathbf{B}\mathbf{y}].$$

Give exact equality conditions in terms of $\mathbf{C} = \mathbf{A} - \mathbf{B}$, $\boldsymbol{\mu}$, and $\boldsymbol{\Sigma}^{1/2}$.

Origin: H Cor. 14.3.11; CLM Lemma 3.2

Solution. Since $\mathbf{C} \succeq \mathbf{0}$,

$$\mathbb{E}[\mathbf{y}'\mathbf{A}\mathbf{y}] - \mathbb{E}[\mathbf{y}'\mathbf{B}\mathbf{y}] = \text{tr}(\mathbf{C}\boldsymbol{\Sigma}) + \boldsymbol{\mu}'\mathbf{C}\boldsymbol{\mu}.$$

Let $\mathbf{C}^{1/2}$ and $\boldsymbol{\Sigma}^{1/2}$ be the symmetric square roots. Then

$$\begin{aligned} \text{tr}(\mathbf{C}\boldsymbol{\Sigma}) &= \text{tr}(\boldsymbol{\Sigma}^{1/2}\mathbf{C}\boldsymbol{\Sigma}^{1/2}) = \text{tr}\{(\mathbf{C}^{1/2}\boldsymbol{\Sigma}^{1/2})'(\mathbf{C}^{1/2}\boldsymbol{\Sigma}^{1/2})\} \geq 0, \\ \boldsymbol{\mu}'\mathbf{C}\boldsymbol{\mu} &= \|\mathbf{C}^{1/2}\boldsymbol{\mu}\|^2 \geq 0. \end{aligned}$$

Equality holds if and only if

$$\mathbf{C}^{1/2}\boldsymbol{\Sigma}^{1/2} = \mathbf{0}, \quad \mathbf{C}^{1/2}\boldsymbol{\mu} = \mathbf{0}.$$

Equivalently, $\mathbf{C}\Sigma^{1/2} = \mathbf{0}$ and $\mathbf{C}\boldsymbol{\mu} = \mathbf{0}$. ■

Exercise 10.14 (A quadratic form with prescribed expectation). Suppose

$$\mathbb{E}[\mathbf{y}] = \mathbf{X}\boldsymbol{\beta}, \quad \text{cov}(\mathbf{y}) = \sigma^2 \mathbf{I}_n,$$

where $\mathbf{X} \in \mathbb{R}^{n \times p}$ is fixed and $\boldsymbol{\beta} \in \mathbb{R}^p$ and $\sigma^2 \geq 0$ are arbitrary. For symmetric constant $\mathbf{A}$, prove

$$\mathbb{E}[\mathbf{y}'\mathbf{A}\mathbf{y}] = \sigma^2 \quad \text{for all } \boldsymbol{\beta}, \sigma^2$$

if and only if

$$\text{tr}(\mathbf{A}) = 1, \quad \mathbf{X}'\mathbf{A}\mathbf{X} = \mathbf{0}.$$

Show that $\mathbf{A} = \mathbf{M}/(n - p)$ satisfies these conditions when $\text{rank}(\mathbf{X}) = p < n$.

Origin: Adapted from CLM Thm. 3.4; A&M Ex. 13.60

Solution. Theorem 10.26 gives

$$\mathbb{E}[\mathbf{y}'\mathbf{A}\mathbf{y}] = \sigma^2 \text{tr}(\mathbf{A}) + \boldsymbol{\beta}'\mathbf{X}'\mathbf{A}\mathbf{X}\boldsymbol{\beta}.$$

If this equals σ^2 for every $\boldsymbol{\beta}, \sigma^2$, set $\boldsymbol{\beta} = \mathbf{0}$ and $\sigma^2 = 1$ to get $\text{tr}(\mathbf{A}) = 1$. Then set $\sigma^2 = 0$:

$$\boldsymbol{\beta}'\mathbf{X}'\mathbf{A}\mathbf{X}\boldsymbol{\beta} = 0 \quad \forall \boldsymbol{\beta}.$$

Theorem 10.13 gives $\mathbf{X}'\mathbf{A}\mathbf{X} = \mathbf{0}$. The converse follows by substitution.
For $\mathbf{M} = \mathbf{I}_n - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$,

$$\mathbf{M}\mathbf{X} = \mathbf{0}, \quad \text{tr} \left(\frac{\mathbf{M}}{n-p} \right) = \frac{\text{rank}(\mathbf{M})}{n-p} = 1.$$



Exercise 10.15 (Idempotent Gram matrices). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Prove

$$\mathbf{A}'\mathbf{A} \text{ is idempotent} \iff \mathbf{A}\mathbf{A}' \text{ is idempotent.}$$

Under either condition, prove

$$\mathbf{A} = \mathbf{A}\mathbf{A}'\mathbf{A}.$$

Origin: Adapted from H Ex. 10.4

Solution. Suppose $\mathbf{C} = \mathbf{A}'\mathbf{A}$ is idempotent. Since $\mathbf{C}' = \mathbf{C}$,

$$\|\mathbf{A}(\mathbf{I}_n - \mathbf{C})\mathbf{x}\|^2 = \mathbf{x}'(\mathbf{I}_n - \mathbf{C})\mathbf{C}(\mathbf{I}_n - \mathbf{C})\mathbf{x} = 0 \quad (\mathbf{x} \in \mathbb{R}^n).$$

Thus $\mathbf{A}(\mathbf{I}_n - \mathbf{C}) = \mathbf{O}$, and hence

$$\mathbf{A} = \mathbf{A}\mathbf{C} = \mathbf{A}\mathbf{A}'\mathbf{A}, \quad (\mathbf{A}\mathbf{A}')^2 = \mathbf{A}(\mathbf{A}'\mathbf{A})\mathbf{A}' = \mathbf{A}\mathbf{A}'.$$

Applying this implication to $\mathbf{A}'$ proves the converse and the same identity. 